Artemy Kolchinsky

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EDUCATION

Indiana University (Bloomington, IN, USA), 2015
Ph.D. in Informatics (focus in Complex Systems), Minor in Cognitive Science
Thesis: "Measuring Scales: Integration and Modularity in Complex Systems"
Committee: Luis M. Rocha (chair), Yong-Yeol Ahn, Randall Beer, Alessandro Flammini, Olaf Sporns

New York University (New York, NY, USA), 2004
B.A. Magna Cum Laude, Individualized Study (concentration in Complex Systems)

ACADEMIC POSITIONS

6/2023-Present **Universitat Pompeu Fabra** (Barcelona, Spain)
Marie Curie postdoctoral fellow

1/2022-5/2023 **University of Tokyo** (Tokyo, Japan)
Project researcher at the Universal Biology Institute

12/2015-7/2021 **Santa Fe Institute** (Santa Fe, NM, USA)
Program postdoctoral fellow with advisor David H. Wolpert

INDUSTRY

Summer 2014 **LinkedIn Corporation** (Mountain View, CA, USA)
Data science internship

PUBLICATIONS

A. Kolchinsky, "Thermodynamics of Darwinian evolution in molecular replicators", *Philosophical Transactions of the Royal Society B*, 2025. pdf link

K. R. Thomsen, **A. Kolchinsky**, S. Rasmussen, "Protocellular energy transduction, information, and fitness", *Philosophical Transactions of the Royal Society B*, 2025. pdf link

S. Bartlett, A. W. Eckford, M. Egbert, M. Lingam, **A. Kolchinsky**, A. Frank, G. Ghoshal, "The Physics of Life: Exploring Information as a Distinctive Feature of Living Systems", *PRX Life*, 2025. pdf link

R. Nagayama, K. Yoshimura, **A. Kolchinsky**, S. Ito, "Geometric thermodynamics of reaction-diffusion systems: Thermodynamic trade-off relations and optimal transport for pattern formation", *Physical Review Research*, 2025. pdf link

A. Kolchinsky, I. Marvian, C. Gokler, Z. Liu, P. Shor, O. Shtanko, K. Thompson, D. Wolpert, S. Lloyd, "Maximizing free energy gain", *Entropy*, 2025. pdf link

A. Kolchinsky, "Thermodynamic dissipation does not bound replicator growth and decay rates", *The Journal of Chemical Physics*, 2024. pdf link (2024 JCP Emerging Investigators Special Collection)

R. Solé, C. P. Kempes, B. Corominas-Murtra, M. De Domenico, **A. Kolchinsky**, M. Lachmann, E. Libby, S. Saavedra, E., D. Wolpert, "Fundamental constraints to the logic of living systems", *Royal Society Interface*, 2024. pdf link

P.M. Riechers, C. Gupta, **A. Kolchinsky**, M. Gu, "Thermodynamically ideal quantum state inputs to any device", *PRX Quantum*, 2024. pdf link

A. Kolchinsky, "Partial information decomposition: redundancy as information bottleneck", *Entropy*, 2024. pdf link

J. Piñero, R. Solé, **A. Kolchinsky**, "Optimization of nonequilibrium free energy harvesting illustrated on bacteriorhodopsin", *Physical Review Research*, 2024. pdf link

A. Kolchinsky, N. Ohga, S. Ito, "Thermodynamic bound on spectral perturbations, with applications to oscillations and relaxation dynamics", *Physical Review Research*, 2024. pdf link

D.R. Sowinski, J. Carroll-Nellenback, R.N. Markwick, J. Piñero, M. Gleiser, **A. Kolchinsky**, G. Ghoshal, A. Frank, "Semantic Information in a model of resource gathering agents", *PRX Life*, 2023. pdf link

A. Kolchinsky, "Generalized Zurek's bound on the cost of an individual classical or quantum computation", *Physical Review E*, 2023. pdf link

M. Aguilera and **A. Kolchinsky**, “Quantifying higher-order entropy production in organized nonequilibrium states”, *Proceedings of the ALIFE 2023*, 2023. pdf link

N. Ohga, S. Ito, **A. Kolchinsky**, “Thermodynamic bound on the asymmetry of cross-correlations”, *Physical Review Letters*, 2023. pdf link (*Editors’ Suggestion*; *Featured in Physics*)

K. Yoshimura, **A. Kolchinsky**, A. Dechant, S. Ito, “Housekeeping and excess entropy production for general nonlinear dynamics”, *Physical Review Research*, 2023. pdf link

F.C. Sheldon, **A. Kolchinsky**, F. Caravelli, “Computational capacity of LRC, memristive, and hybrid reservoirs”, *Physical Review E*, 2022. pdf link

A. Kolchinsky, “A novel approach to the partial information decomposition”, *Entropy*, 2022. pdf code link

A. Kolchinsky and D.H. Wolpert, “Dependence of integrated, instantaneous, and fluctuating entropy production on the initial state in quantum and classical processes”, *Physical Review E*, 2021. pdf link

A. Kolchinsky and D.H. Wolpert, “Work, entropy production, and thermodynamics of information under protocol constraints”, *Physical Review X*, 2021. pdf link

A. Kolchinsky and D.H. Wolpert, “Entropy production given constraints on the energy functions”, *Physical Review E*, 2021. pdf link

A. Kolchinsky and D.H. Wolpert, “Thermodynamic costs of Turing machines”, *Physical Review Research*, 2020. pdf link

D.H. Wolpert and **A. Kolchinsky**, “The thermodynamics of computing with circuits”, *New Journal of Physics*, 2020. pdf link

A. Kolchinsky and B. Corominas-Murtra, “Decomposing information into copying versus transformation”, *Royal Society Interface*, 2020. pdf link

A.M. Saxe, Y. Bansal, J. Dapello, M. Advani, **A. Kolchinsky**, B.D. Tracey, D.D. Cox, “On the information bottleneck theory of deep learning”, *Journal of Statistical Mechanics*, 2019. pdf code link

A. Kolchinsky, B.D. Tracey, D.H. Wolpert, “Nonlinear information bottleneck”, *Entropy*, 2019. (*Entropy* 2021 Best Paper Award) pdf link

A. Berdahl, C. Brelsford, C. De Bacco, M. Dumas, V. Ferdinand, J.A. Grochow, L. Hébert-Dufresne, Y. Kallus, C.P. Kempes, **A. Kolchinsky**, D. B. Larremore, E. Libby, E.A. Power, C.A. Stern, B.D.Tracey, “Dynamics of beneficial epidemics”, *Scientific Reports*, 2019. pdf link

E.A. Hobson, V. Ferdinand, **A. Kolchinsky**, J. Garland, “Rethinking animal social complexity measures with the help of complex systems concepts”, *Animal Behaviour*, 2019. pdf link

A. Kolchinsky, B.D. Tracey, S. Van Kuyk, “Caveats for information bottleneck in deterministic scenarios”, *ICLR*, 2019. pdf code link

D.H. Wolpert, **A. Kolchinsky**, J.A. Owen, “A space–time tradeoff for implementing a function with master equation dynamics”, *Nature Communications*, 2019. pdf link

A. Avena-Koenigsberger, X. Yan, **A. Kolchinsky**, M. van den Heuvel, P. Hagmann, O. Sporns, “A spectrum of routing strategies for brain networks”, *PLoS Computational Biology*, 2019. pdf link

J.A. Owen, **A. Kolchinsky**, D.H. Wolpert, “Number of hidden states needed to physically implement a given conditional distribution”, *New Journal of Physics*, 2019. pdf link (correction)

A. Kolchinsky and D.H. Wolpert, “Semantic information, autonomous agency, and nonequilibrium statistical physics”, *Royal Society Interface Focus*, 2018. pdf code link

A.M. Saxe, Y. Bansal, J. Dapello, M. Advani, **A. Kolchinsky**, B.D. Tracey, D.D. Cox, “On the information bottleneck theory of deep learning”, *ICLR*, 2018. pdf code link

A. Kolchinsky, N. Dhande, K. Park, Y.Y. Ahn, “The Minor Fall, the Major Lift: Inferring emotional valence of musical chords through lyrics”, *Royal Society Open Science*, 2017. pdf data code link

A. Kolchinsky, D.H. Wolpert, “Dependence of dissipation on the initial distribution over states”, *Journal of Statistical Mechanics*, 2017. pdf link

A. Kolchinsky, B.D. Tracey, “Estimating mixture entropy with pairwise distances”, *Entropy*, 2017. (correction) pdf code link

A. Kolchinsky, A.J. Gates, L.M. Rocha, “Modularity and the spread of perturbations in complex dynamical systems," *Physical Review E*, 2015. pdf code link

A. Kolchinsky, A. Lourenço, H. Wu, L. Li, L.M. Rocha, "Extraction of pharmacokinetic evidence of drug-drug interactions from the literature," *PLOS One*, 2015. pdf link

A. Kolchinsky, M.P. van den Heuvel, A. Griffa, P. Hagmann, L.M. Rocha, O. Sporns, J. Goñi, "Multi-scale integration and predictability in resting state brain activity," *Frontiers in Neuroinformatics*, 2014. pdf link

A. Rossi, F.J. Parada, **A. Kolchinsky**, A. Puce, "Neural correlates of apparent motion perception of impoverished facial stimuli I: A comparison of ERP and ERSP activity," *NeuroImage*, 2014. pdf link

A. Kolchinsky, A. Lourenço, L. Li, L.M. Rocha, "Evaluation of linear classifiers on articles containing pharmacokinetic evidence of drug-drug interactions," *Proc Pacific Symposium on Biocomputing*, 2013. pdf link

A. Kolchinsky and L.M. Rocha, "Prediction and modularity in dynamical systems," *Proc of European Conf. on the Synthesis and Simulation of Living Systems (ECAL)*, 2011. pdf link

A. Kolchinsky, A. Abi-Haidar, J. Kaur, A.A. Hamed, L.M. Rocha, "Classification of protein-protein interaction full-text documents using text and citation network features," *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2010. pdf link

PREPRINTS

J. Bauermann, G. Bartolucci, **A. Kolchinsky**, "Spatiotemporal organization of chemical oscillators via phase separation", arXiv:2507.16030, 2025. arxiv

G.-H. Xu, **A. Kolchinsky**, J.-C. Delvenne, and S. Ito, "Thermodynamic geometric constraint on the spectrum of Markov rate matrices", arXiv:2507.08938, 2025. arxiv

J. Piñero, **A. Kolchinsky**, S. Redner, R. Solé, "Neutral theory of cooperators", arXiv:2506.09737, 2025. arxiv

M. Aguilera, S. Ito, **A. Kolchinsky**, "Inferring entropy production in many-body systems using nonequilibrium MaxEnt", arXiv:2505.10444, 2025. arxiv

J. Piñero, D. R. Sowinski, G. Ghoshal, A. Frank, **A. Kolchinsky**, "Information bounds production in replicator systems", arXiv:2501.00396, 2024. arxiv.

A. Kolchinsky, A. Dechant, K. Yoshimura, S. Ito, "Generalized free energy and excess entropy production for active systems", arXiv:2412.08432, 2024. arxiv

A. Kolchinsky, A. Dechant, K. Yoshimura, S. Ito, "Information geometry of excess and housekeeping entropy production", arXiv:2206.14599, 2022. arxiv

C. Gokler, **A. Kolchinsky**, Z. Liu, I. Marvian , P. Shor, O. Shtanko, K. Thompson, D. Wolpert, S. Lloyd, "When is a bit worth much more than $kT \ln 2$?", arXiv:1705.09598, 2017. arxiv

INVITED TALKS

11/2025 *Kyoto Workshop on Quantum Thermodynamics & Stochastic Thermodynamics*, Kyoto University, Japan
10/2025 *Autocatalysis in Reaction Networks Seminar (ARN)* (online)
9/2025 *ILIAD2: ODYSSEY, conference on mathematical alignment*, Berkeley, CA, USA
6/2025 *CRC Colloquium*, Free University of Berlin, Berlin, Germany
5/2025 *Joint European Thermodynamics Conference*, Belgrade, Serbia
11/2024 *Universal Biology Institute Seminar Series*, University of Tokyo, Japan
9/2024 *Information in Matter Colloquium*, AMOLF, Amsterdam, Netherlands
7/2024 *Physics Department Seminar*, University of Barcelona, Barcelona, Spain
2/2024 *Seminar*, Barcelona Collaboratorium for Modelling and Predictive Biology, Barcelona, Spain
11/2023 *Mini-workshop: Non-equilibrium thermodynamics and biology*, Biofiska Institute, Bilbao, Spain
11/2023 *Seminar*, Basque Center for Applied Mathematics, Bilbao, Spain
7/2023 *Information Engines at the Frontiers of Nanoscale Thermodynamics*, Telluride, CO, USA
6/2023 *Webinar: Mathematics, Physics & Machine Learning*, Instituto Superior Técnico, Lisbon, Portugal
6/2023 *Conference: Decomposing Multivariate Information in Complex Systems*
Max Planck Institute for the Physics of Complex Systems, Dresden, Germany
9/2022 *Seminar*, Dutch Institute of Emergent Phenomena, University of Amsterdam, Netherlands
8/2022 *Seminar*, Earth-Life Science Institute (ELSI), Tokyo, Japan
6/2022 *Evolution of complexity from the statistical physics perspective*, Yerevan Physics Institute, Armenia

6/2022	*Yagami Statistical Physics Seminar*, Keio University, Japan
5/2022	*Workshop on Stochastic Thermodynamics III*, University of Tokyo, Japan
4/2022	*Cross Labs Workshop: Is AI Extending the Mind?*, Cross Labs, Japan
2/2022	*Physics Department Colloquium Series*, University of Rochester, NY, USA
12/2021	*Universal Biology Institute Seminar Series*, University of Tokyo, Japan
02/2021	*Workshop: Origins of Life: The Possible and the Actual*, Santa Fe Institute, NM, USA
7/2020	*ICTP Seminar Series*, Abdus Salam International Center for Theoretical Physics, Trieste, Italy
2/2020	*AI Seminar Series*, Information Sciences Institute, Los Angeles, CA, USA
7/2019	*ISTI Seminar Series*, Los Alamos National Lab, Los Alamos, NM, USA
6/2018	*Connectomics Lecture Series*, Universidad Diego Portales, Santiago, Chile
4/2018	*Meeting of the Society for the Neural Control of Movement*, Santa Fe, NM, USA
11/2017	Seoul National University, South Korea
8/2017	*Workshop: Thermodynamics & Computation: Towards a New Synthesis*, Santa Fe Institute, NM, USA
10/2016	*Statistical Physics, Information Processing and Biology*, Santa Fe Institute, NM, USA
2/2016	Information Sciences Institute, Los Angeles, CA, USA

Awards & Fellowships

2022 Marie Curie Individual Fellowship (HORIZON-MSCA-2021-PF-01, #101068029)
"NETOLIFE: Nonequilibrium thermodynamics of the origin of life"

Grants

2023 John Templeton Foundation (#62828), $947,926, Co-PI
"Goal-directed behavior and the origin of life"

2022 John Templeton Foundation (#62417), $627,227, Co-PI
"Information architectures that enable life: the emergence of meaning"

2019 Foundational Questions Institute (FQXi-RFP-IPW-1912), $118,100, Co-Investigator (PI: David Wolpert)
"The role of constraints in the thermodynamics of intelligence"

2016 Foundational Questions Institute (FQXi-RFP-1622), $128,319, Co-Investigator (PI: David Wolpert)
"Observers as self-maintaining non-equilibrium systems"

2016 NSF INSPIRE (CHE-1648973), $999,947, co-author (PI: David Wolpert)
"Tradeoffs in the thermodynamics of computation: a new paradigm for biological information-processing"

2016 NSF (1620462), $770,000, co-author (PI: David Wolpert)
"Information Networks and the Evolution of Social Organizations"

Popular Science & Outreach

Articles

A. Kolchinsky, "The cost of sending a bit across a living cell", *Physics Magazine*, 2023. link

Public talks

4/2018 *SITE Santa Fe* (contemporary art museum), Santa Fe, NM, USA
Public talk on "Life, entropy, and the 2^{nd} law of thermodynamics"

Teaching

Online courses

2018 "Fundamentals of machine learning" (w/ B.D. Tracey), Santa Fe Institute *Complexity Explorer*
Designed and delivered a 12-part online course on basics of machine learning (link)

Workshops, lectures, and tutorials

6/2019 Santa Fe Institute Complex Systems Summer School, NM, USA
Two-part lecture series on machine learning and its research frontiers

3/2019 Santa Fe Institute, NM, USA
Tutorial on "Machine learning with TensorFlow"

6/2017, 6/2018 Santa Fe Institute, NM, USA
Tutorial on introduction to programming and data analysis in Python (w/ V. Ferdinand)

11/2017 Seoul National University, Seoul, South Korea
Week-long workshop organized w/ V. Ferdinand
"Thermodynamics, evolution, and inference through the lens of information theory"

11/2017 ACtioN/Trustee Meeting, Santa Fe Institute, NM, USA
Lecture on "Machine learning: A guide for the perplexed" (w/ B. D. Tracey)

5/2010 Instituto Gulbenkian de Ciência, Oeiras, Portugal
Assisted with week-long "Bayesian brain" educational module

Teaching Assistant at Indiana University, Bloomington, IN

Spring 2014 "I400 Large-scale Social Phenomena" with Simon DeDeo (link)

Spring 2011 "I201 Math and logic foundations of Informatics" with Erik Wennstrom

Fall 2010 "I485 Biologically Inspired Computing" (link) with Luis M. Rocha
Developed and implemented a 5-part problem set for a computational lab

Fall 2008-Spring 2009 "I210 Information Infrastructure" (various instructors)
Ran lab to assist with Pyton programming course

ADVISING

PhD dissertation committees

2025 *André Gomes*, PhD, Electrical & Computer Engineering, Instituto Superior Técnico, Lisbon, Portugal
Thesis: "Advances in Partial Information Decomposition"

Undergraduate projects supervised

2018 *Nicolas Freitas*, Santa Fe Institute Summer REU Program, Santa Fe, NM, USA
Project: "Scaling of information in biochemical systems"

2017 *Francis Cavanna*, Santa Fe Institute Summer REU Program, Santa Fe, NM, USA
Project: "Investigating the relationship between criticality and Landauer costs using the Ising model"

ACADEMIC SERVICE

Conference Organization

12/2025 Kyoto Workshop on Quantum Thermodynamics and Stochastic Thermodynamics, co-organizer
Kyoto University, Japan (link)

7/2024 Information-Driven States of Matter, co-organizer
University of Rochester, NY, USA (pdf program) press

Review Panels

2025 Interdisciplinary Assessment Board for Artificial Intelligence, Spanish Research Agency, reviewer

2021- John Templeton Foundation, external reviewer

2023 Swiss National Science Foundation, external reviewer

2020 European Research Council (ERC), Starting Grant Panel, external reviewer

Guest editor

Entropy special issue on "Thermodynamics and Information Theory of Living Systems" link

Reviewer

General: *PNAS*, *Nature Communications*, *npj Complexity*, *Interface Focus*
Physics: *PRL*, *PRX*, *PRX Quantum*, *PRR*, *PRE*, *NJP*, *Frontiers in Physics*, *J of Physics A*, *Physica A*, *EPL*
Biology: *Proc R Soc B*, *Phil Trans B*, *PLoS Comp Bio*, *Theory in the Biosciences*, *BioSystems*
Information theory and machine learning: *Entropy*, *ICLR*, *IEEE TPAMI*, *Kybernetika*, *Applied Sciences*

Other

2008-2013 Organized a weekly discussion group on complexity, dynamical systems, and embodiment in cognitive science, Indiana University, Bloomington, IN (link)

SKILLS

Programming: Python, MATLAB, C, C++, R, Java
Scientific computing, machine learning, web programming, databases/SQL

Languages: Fluent: English, Russian, Spanish. Basic: Portuguese.